

Figure 7. Prompt-range M2 phase diagrams

Draft multipanel assembly from audited CSV/PNG assets.

A Strict equal-frame winners

M2 strict equal frame non-oracle winners		rho=0.01		rho=0.03		rho=0.1		rho=0.3		rho=1		rho=3		rho=10	
sigma=0.05	srht	srht	srht	srht	srht	srht	srht	srht	srht	srht	srht	srht	srht	srht	srht
	pernoise	pernoise	pernoise	pernoise	pernoise	pernoise	pernoise	pernoise	pernoise	pernoise	pernoise	pernoise	pernoise	pernoise	pernoise
	44.7 dB	39.9 dB	34.7 dB	30.0 dB	25.0 dB	21.0 dB	17.7 dB	16.3 dB	16.2 dB						
sigma=0.1	srht	srht	srht	srht	srht	srht	srht	sub-floor	sub-floor	sub-floor	sub-floor	sub-floor	sub-floor	sub-floor	sub-floor
	pernoise	pernoise	pernoise	pernoise	pernoise	pernoise	pernoise	noise	noise	noise	noise	noise	noise	noise	noise
	38.7 dB	33.9 dB	28.9 dB	24.7 dB	19.5 dB	16.0 dB	13.7 dB	12.9 dB	12.9 dB						
sigma=0.15	srht	srht	srht	srht	srht	hadamard_jordan	sub-floor	sub-floor	sub-floor	sub-floor	sub-floor	sub-floor	sub-floor	sub-floor	sub-floor
	pernoise	pernoise	pernoise	pernoise	pernoise	sgc_proxy	noise	noise	noise	noise	noise	noise	noise	noise	noise
	35.3 dB	30.4 dB	25.3 dB	20.8 dB	16.8 dB	13.8 dB	12.3 dB	11.8 dB	11.8 dB						
sigma=0.3	srht	srht	srht	srht	srht	sub-floor	sub-floor	sub-floor	sub-floor	sub-floor	sub-floor	sub-floor	sub-floor	sub-floor	sub-floor
	pernoise	pernoise	pernoise	pernoise	pernoise	noise	noise	noise	noise	noise	noise	noise	noise	noise	noise
	29.2 dB	24.6 dB	19.8 dB	16.0 dB	13.1 dB	11.1 dB	10.9 dB	10.9 dB	10.9 dB						
sigma=0.5	srht	srht	srht	sub-floor	sub-floor	sub-floor	sub-floor	sub-floor	sub-floor	sub-floor	sub-floor	sub-floor	sub-floor	sub-floor	sub-floor
	pernoise	pernoise	pernoise	noise	noise	noise	noise	noise	noise	noise	noise	noise	noise	noise	noise
	24.9 dB	20.4 dB	16.2 dB	13.4 dB	11.7 dB	11.0 dB	10.9 dB	10.9 dB	10.7 dB						

Grey cells are sub-floor no reconstruction-quality winners acknowledged

B All-non-oracle winners

M2 all-non-oracle winners with reference overhead		rho=0.01		rho=0.03		rho=0.1		rho=0.3		rho=1		rho=3		rho=10	
sigma=0.05	srht	srht	srht	srht	srht	srht	srht	srht	srht	srht	srht	srht	srht	srht	srht
	ref2	ref2	ref2	ref2	ref2	ref2	ref2	ref2	ref2	ref2	ref2	ref2	ref2	ref2	ref2
	47.7 dB	42.9 dB	37.7 dB	32.9 dB	27.8 dB	23.3 dB	19.3 dB	17.3 dB	17.3 dB						
sigma=0.1	srht	srht	srht	srht	srht	srht	sub-floor	sub-floor	sub-floor	sub-floor	sub-floor	sub-floor	sub-floor	sub-floor	sub-floor
	ref2	ref2	ref2	ref2	ref2	ref2	noise	noise	noise	noise	noise	noise	noise	noise	noise
	41.7 dB	36.9 dB	31.7 dB	27.6 dB	22.6 dB	17.9 dB	14.7 dB	13.4 dB	13.4 dB						
sigma=0.15	srht	srht	srht	srht	srht	srht	sub-floor	sub-floor	sub-floor	sub-floor	sub-floor	sub-floor	sub-floor	sub-floor	sub-floor
	ref2	ref2	ref2	ref2	ref2	ref2	noise	noise	noise	noise	noise	noise	noise	noise	noise
	38.1 dB	33.4 dB	28.2 dB	23.6 dB	18.9 dB	16.3 dB	12.9 dB	12.0 dB	12.0 dB						
sigma=0.3	srht	srht	srht	srht	srht	srht	sub-floor	sub-floor	sub-floor	sub-floor	sub-floor	sub-floor	sub-floor	sub-floor	sub-floor
	ref2	ref2	ref2	ref2	ref2	ref2	noise	noise	noise	noise	noise	noise	noise	noise	noise
	32.2 dB	27.5 dB	22.4 dB	18.3 dB	14.6 dB	12.3 dB	11.3 dB	11.0 dB	10.9 dB						
sigma=0.5	srht	srht	srht	srht	sub-floor	sub-floor	sub-floor	sub-floor	sub-floor	sub-floor	sub-floor	sub-floor	sub-floor	sub-floor	sub-floor
	ref2	ref2	ref2	ref2	noise	noise	noise	noise	noise	noise	noise	noise	noise	noise	noise
	27.8 dB	23.2 dB	18.5 dB	16.9 dB	16.9 dB	12.4 dB	11.3 dB	10.8 dB	10.7 dB						

Grey cells are sub-floor no reconstruction-quality winners acknowledged

C Observed boundary fits

M2 observed log-rho boundary fits

correction	challenger	baseline	observed_points	sigma_a_exponent	r2	rmse_log10
reference_k8	srht_paired	hadamard_paired	5	-1.9285	0.99951	0.015015
none	srht_paired	hadamard_paired	3	-0.94178	0.99213	0.016523
reference_k32	srht_paired	hadamard_paired	3	-1.0962	0.9916	0.019875
sgc_proxy	srht_paired	hadamard_paired	5	-1.5899	0.9889	0.05916

Strict equal-frame and all-non-oracle maps cover rho=0.001..10 and sigma_a=0.05..0.50; observed boundary fits summarize the resolved flip curves.

Resolution: 300 dpi; source assets listed in paper_multipanel_manifest.csv.